

Robocup - Final Design Report

Team 28

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1 Executive Summary

This report explores the designed robot, nicknamed "Sucky Sucky 9000", and its performance in the Robocup competition and against fellow competitor robots. The robot, utilising circular geometry, a compact harvester, weight storage system, and a streamlined form of SLAM computer vision, was able to win four rounds, losing two and placing 7th overall in the competition.

Adjustments were made to the mechanical design throughout the rounds, utilising a smoother ramp design and new guide walls with greater coverage to fix identified issues from previous rounds. Further improvements were made to the revolver sorter drum and its door, as well as sensor placement and modification to the MDF levels for weight collection and wire organisation. The software's SLAM implementation was improved through using the Theta* path finding algorithm, increasing the speed of navigation and a weighted search algorithm which enhanced the robot's searching capabilities.

Three competitors were chosen to be compared against over the course of the 6-round duration, group 12, group 26, and group 39. Group 12's robot was found to be the winner of the competition, with strengths in all major categories due to the simplistic design and considerable testing time undertaken by that team. Group 26's robot had significant reliability issues through its harvester system, navigation, and weight sorting system. This was a massive performance change when compared against the reasonable testing data and pre-competition evaluation of the design. Group 39's robot had some issues with navigation and general sensing, however its' weight collection and sorting system was far more effective and efficient compared to the current design, and was the robot to knock it out in the competition.

The issues could be pointed to two specific faults as pointed out in the Fault Tree Analysis (FTA), the weight jamming in the sorter and the robot being continually stuck for a significant amount of time. The weight jamming was the most significant issue, which was due to the factors of the ramp, as well as the non-alignment of the sorter when it tried to collect a weight.

The mechanical adjustments that were recommended for the design were the use of a Geneva mechanism that would locate the drum far more accurately than the current single motor design. A watchdog timer was recommended on the software to check change in IMU position, detecting if the robot became stuck during the competition.

2 Introduction

The Robocup project revolved around creating a robot design that would be able to detect, collect and sort various weights as it travels around an arena avoiding obstacles. It was deemed that the robot needed a navigation system, traversal system, and a weight collection and sorting system designed, along with a base structure to house these features. In the conceptual stage, it was decided to pursue a circular structure design to geometrically solve the problem of the robot getting stuck.

This choice created some restrictions on the design of physical systems. The traversal system was a tensioned track system built into the circular structure. The weight collection used a harvester system paired with a custom ramp to guide the weights to the sorter, which was a revolving drum. Finally the navigation system was the most complicated, utilising an efficient form of SLAM computer vision to guide the robot around the arena for the competition.

This report shows and discusses the performance of the design and its core features during the competition when compared against the testing done prior. Any mid-competition changes will be covered, as well as a comparison against 3 other competitor robots to evaluate strengths, weaknesses and possible future improvements to the design.

3 Design Description

3.1 Mechanical Design

The mechanical design of the robot is split into 3 sections; the body structure, traction system, and the weight collection/sorting system. The body was a circular design, utilizing MDF boards and aluminium profile beams to create a 3-level system to compartmentalize different components of the robot. The traction system relied on tracks internal to the robot, which used a tensioning system to mechanically tune the track drift of the robot. The final design used within competition is displayed in Figure 1.

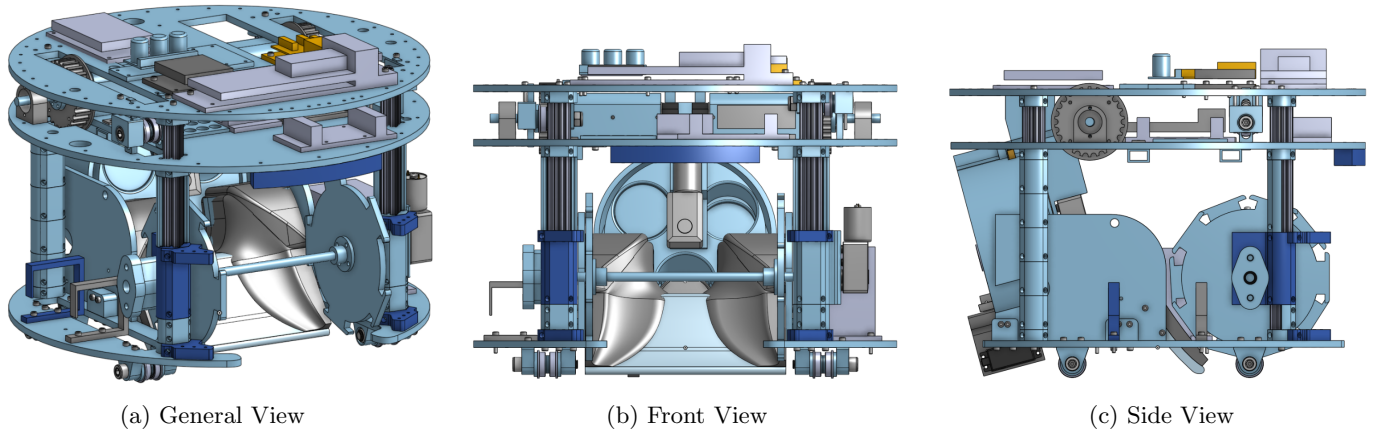


Figure 1: Modelled version of the final design used in competition.

Finally there is the weight collection/sorting system. Initially there is a harvester wheel, which is built into the robot's frame, that is powered by a 50 RPM motor which pulls the weights up a ramp. The ramp uses guide walls which shifts the weight into the revolving sorting system. This system then uses a limit switch on a back door of the revolver, which detects a weight entered the sorter. An inductive sensor, on the side of the sorter, detects if weights are metal or not. It will either eject the weight (in the case of a plastic weight) or will rotate the drum using a 10 RPM motor (in the case of a metal weight). A maximum of two weights were able to be held on board of the robot, if two weights were picked up the robot would return to base to then rotate the drum to drop them off. This logic can be shown by a high-level function block in Figure 2 below.

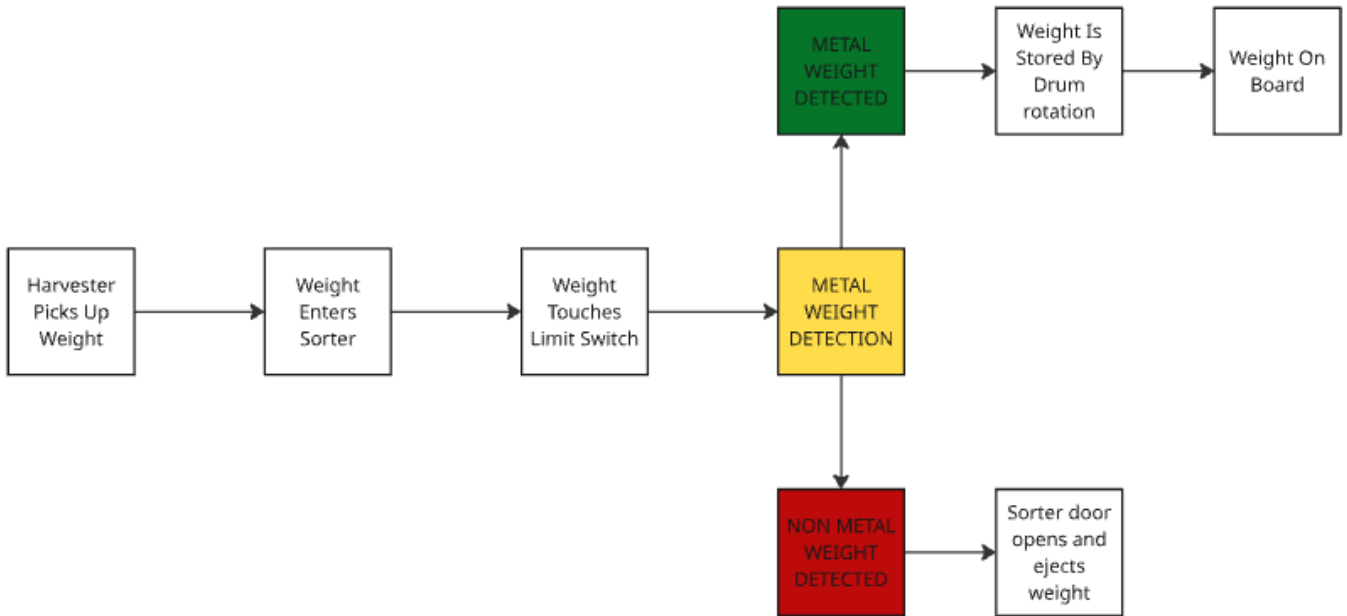


Figure 2: High-Level robot sorting logic for mechanical components to be designed for

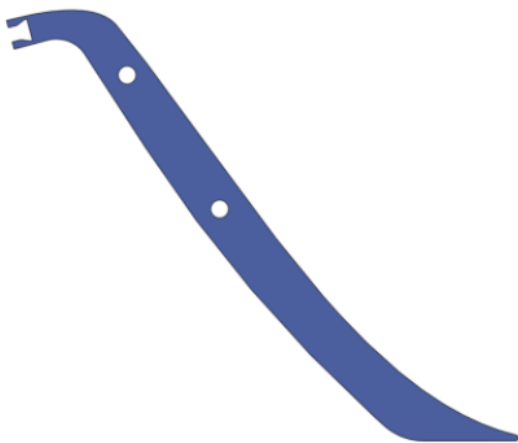
3.2 Mechanical Design Modifications

Several modifications to the mechanical design were made directly before the competition or during the down time between matches. Such changes were as follows:

3.2.1 Ramp wall modifications

The ramp and it's walls were improved after the first day of the competition due to jamming issues. The new ramp changed out the sharp initial circular path for instead a smooth linear-like ramp design as shown in Figure 3. This design was found to pull up weights in all orientations and avoiding previous jamming issues. The ramp walls were also

OLD DESIGN:



NEW DESIGN:

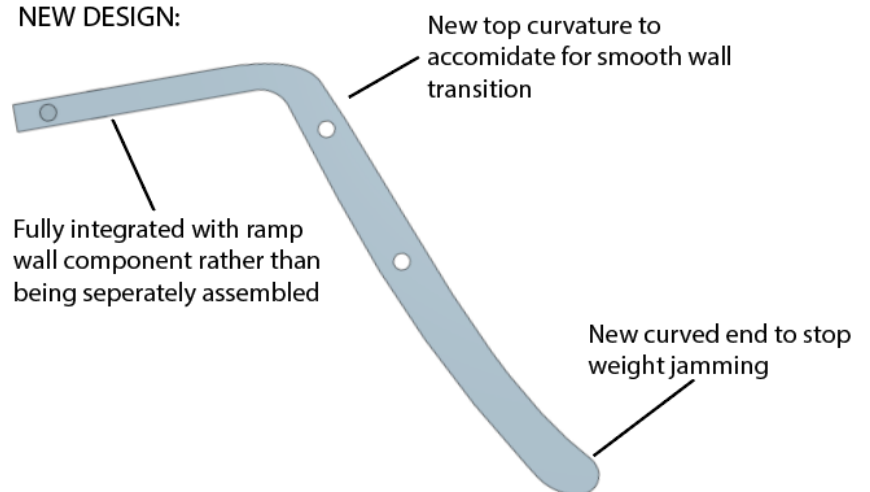


Figure 3: Comparison between old and new ramp designs with commentary on specific design changes

remodeled to improve the weight guiding capability. The initial wall design guided the weight at the end of ramp. The improved walls used inspiration from a pair of clasping hands to create walls that gradually moved the weight towards the center of the ramp in the correct orientation. This design is shown in 4 This modification was found to significantly improve weight pickup success and depositing into the sorter. However another issue found was that the weight processing time was still slow enough to have the potential issue of capturing two weights and then jamming the system.

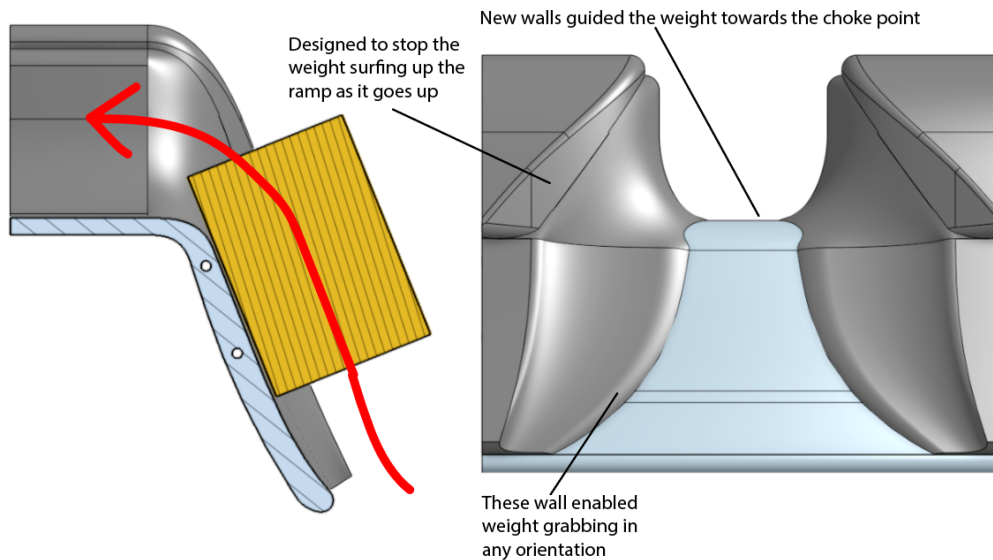


Figure 4: Ramp Wall changes and logic breakdown

3.2.2 Revolver sorter modifications

The revolver sorter door in the initial design had a simple limit switch on the edge of the door to detect all weights. This method however proved unreliable during testing. A solution was made where the limit switch was moved centrally in the door and a button was developed using a loose 3D printed component. The design is shown in Figure 5.

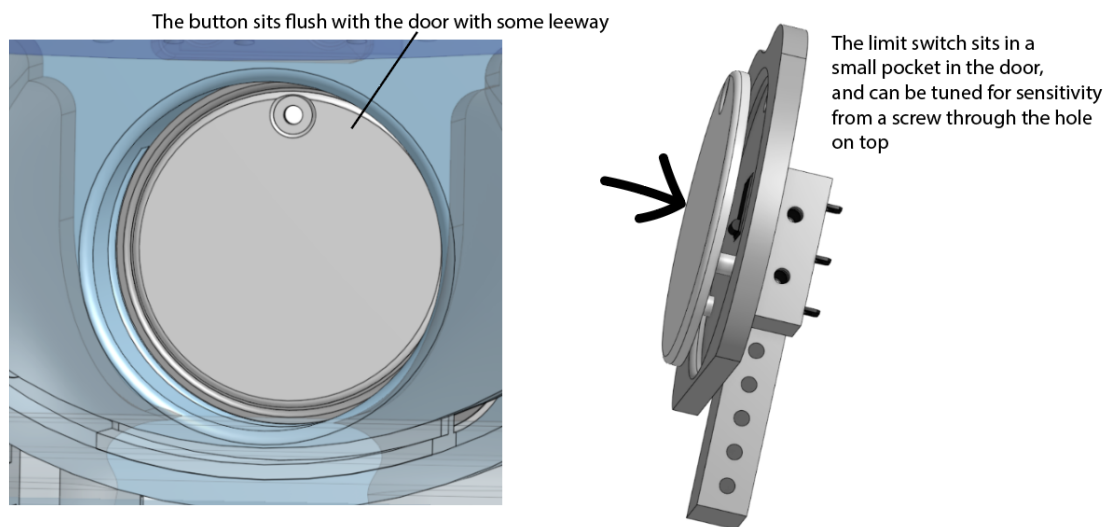


Figure 5: Break down of door operations

This design turned the whole area into a button sensor, improving the weight detection effectiveness. The door motor was also upgraded to the HX 10 as the previous motor could not withstand the impulse of the weights hitting the door.

The revolver drum had a mounting modification upgrade to the internal structure, as it was found a heavy weight would cause the drum to slip from the motor position. This involved boring a hole out of the connecting rod to thread a screw through. Then the revolver sorter had a new hole modelled that had a variable range to allow for easier construction. The new modifications improved the movement accuracy of the drum, which improving overall sorting efficiency. This new design for the drum can be shown in Figure 6

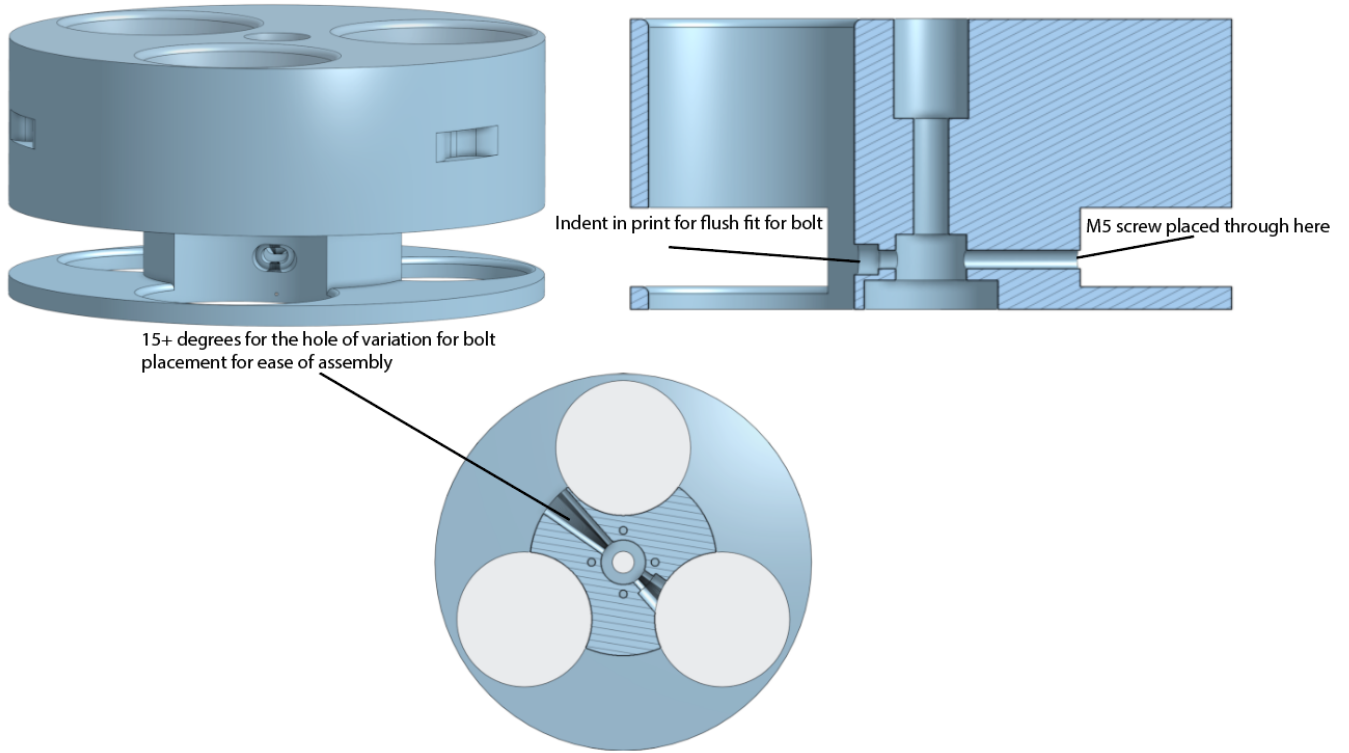
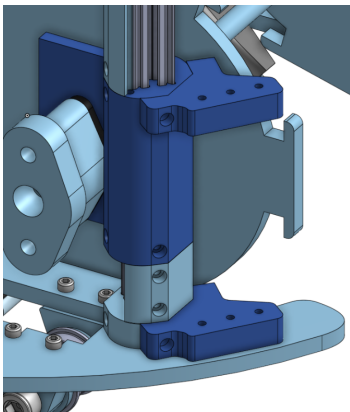


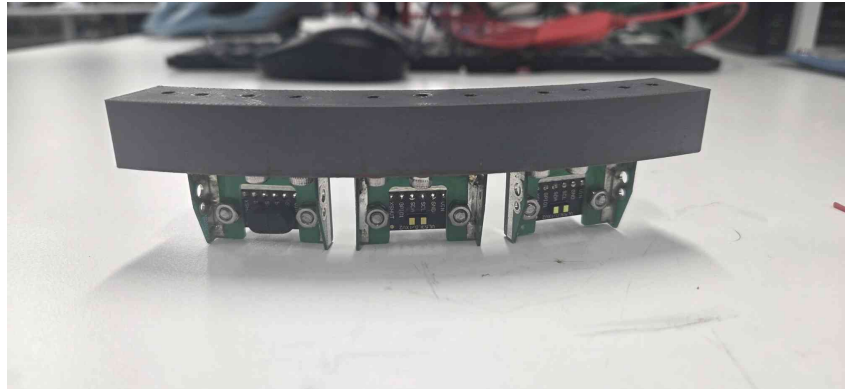
Figure 6: Sorting drum modification with connecting rod.

3.2.3 Structures to hold sensing array

New sensors were mounted onto the robot's support columns, using a 3D printed mount that allows the Time Of Flight (TOF) sensors to be mounted upon the columns at 37.5 degrees from the front. These were placed at different heights for weight differentiation, which can be seen in Figure 7a. Another array was placed on the bottom side of the second level to provide better Field of View (FOV) coverage for the software, which is shown in Figure 7b.



(a) Side column TOF sensor mounts



(b) Main TOF sensor mounts

Figure 7: Sensor modifications added for extra data

This simple mount utilized existing holes on the robot's previous sensor array attempt to mount on a 3D printed part with 3 TOF sensor spaces.

3.2.4 Construction improvements

The base level of the robot, originally the opening for the harvester was quite sharp, and left the possibility of the weight hitting the harvester and damaging the system. The new bottom level addresses this issue through creating an appendage that follows the curvature, which then allows for a smoother transition for the weight and secures the harvester, as shown in Figure 8.

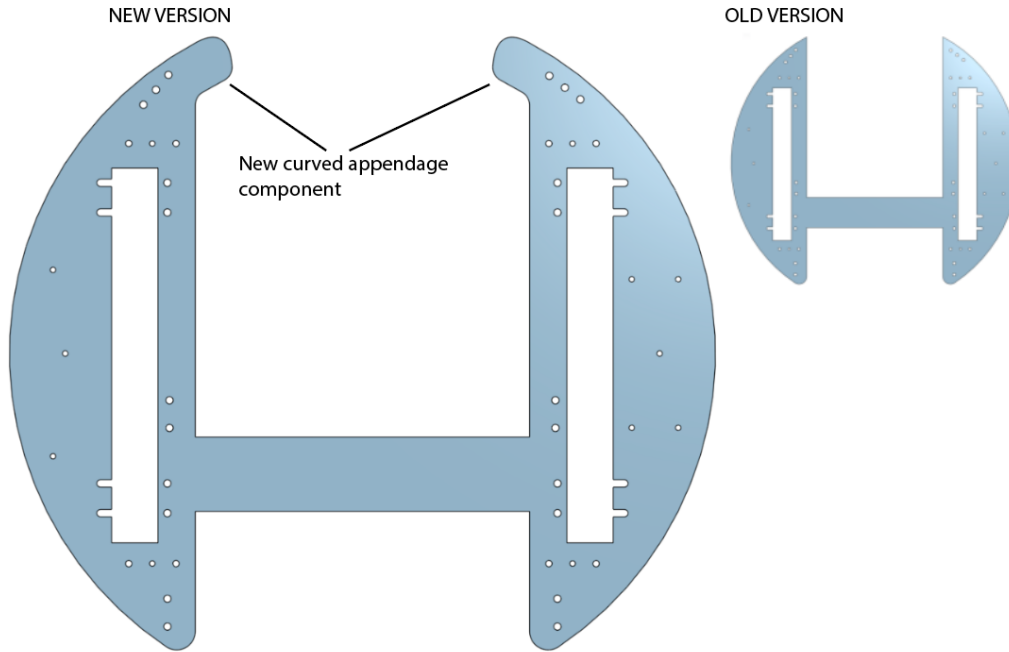


Figure 8: Level 1 modification highlighted and compared to the old version

The final improvement was made to incorporating 3D printed parts to help with wire management, as that was seen as a major issue to the robot's endurance in the competition. These were included on the bottom of the second level to route the important power wires, and further cuts were made in the MDF to allow for easier routing.

3.3 Electrical Design

The electrical design involved the battery and power supply being held at the bottom level, with direct routing between each other and the boards on the higher levels. The 2nd level housed the PPM motor drives for both the harvester, sorter motor and drive motors, as well as the inductive sensor boards. The top layer contained the CPU, encoder board, and digital level shift to co-ordinate the decisions and signal outputs to the rest of the system.

3.4 Electrical Design Modifications

The main improvement was on cable management and cable quality across the system. The door limit switch wiring was changed to use a thicker wire due to the stress from slamming the door on the ground damaging the connection. Both limit switches for the door and sorter were wrapped in extra insulation to make identification easy, and to protect from potential damage. This wiring was pointed out in Figure 9.

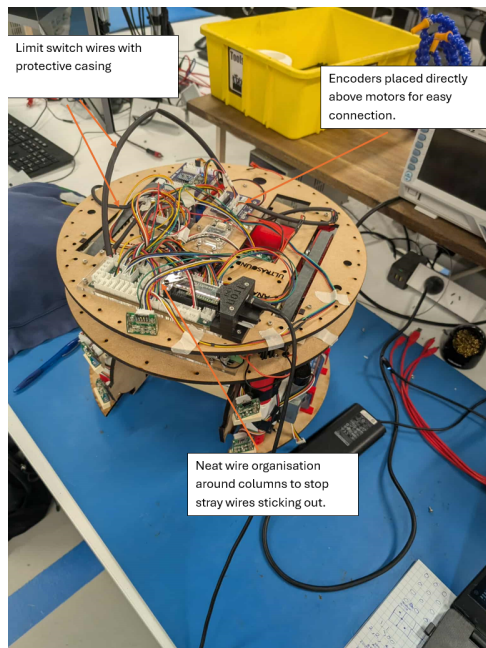


Figure 9: Example of wiring with points of interest highlighted

The cable management utilized set routes within the MDF to guide the wires through the robot, avoiding exposure on the robot's outside and possibility for damage. Tape was used to keep the wiring tight to the robot, and labeling on the wires allowed for easy diagnosis and rewiring when removing or replacing components. The electrical boards were placed in logical areas to optimize this organization and natural wire flow.

3.5 Software Design

The software had three main functionalities; pose, object detection, and pathing. Each feature utilised the physical circular design to allow for accurate and effective functionality. Pose was determined using encoders attached to the motors that drove the tracks, alongside this the IMU was used to determine the rotation of the robot. At each loop of the software, the position was updated by multiplying the average change in encoder positions and then converting this to X and Y displacement using the IMU angle.

The main high level state machine is presented in Figure 10 demonstrating a random walk. In software a queue was used which represented positions the robot would need to travel to. If the end of the queue was reached then a new location was generated randomly.

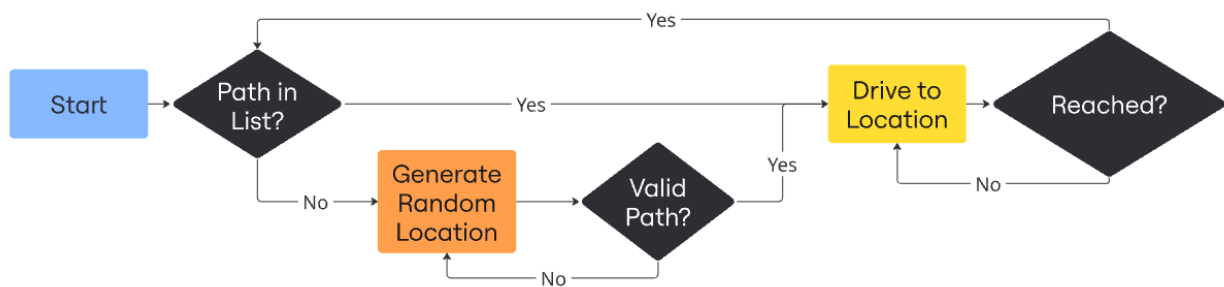


Figure 10: State diagram of overall robot pathing.

Object detection used a 48x98 probability matrix with each cell representing a confidence value of whether the cell contained a wall or free space. Five TOF sensors at the front of the robot were used with a ray-tracing algorithm to adjust values inside the matrix. If the sensors detected anything the state diagram in 11 was used.

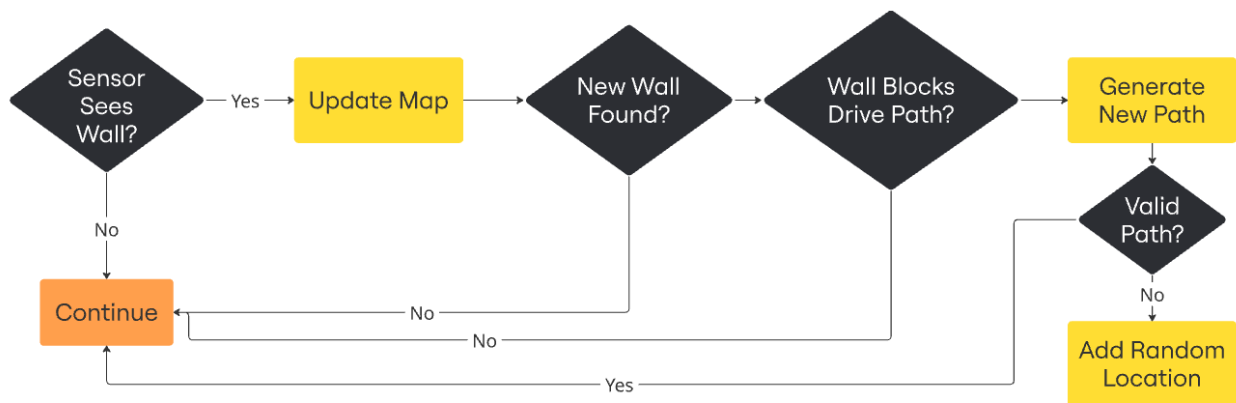


Figure 11: State diagram of sensors updating the map.

As shown in the state diagram a "wall" was updated into the map if a TOF sensor detected it, and any intermediate cells were updated as "free space". If a new wall that hadn't previously been seen was detected, the location was checked against the current path to see if there was an interference. If an interference occurred a new path was generated, if no valid path could be found a new location was chosen to travel to.

The final software element was weight detection, two short TOF sensors were placed at the bottom of the robot level with the weights. If the top TOF sensors detected something different in comparison to the bottom TOF sensors a weight was assumed to be at the location detected by the bottom sensors and the state diagram in 12 was started.

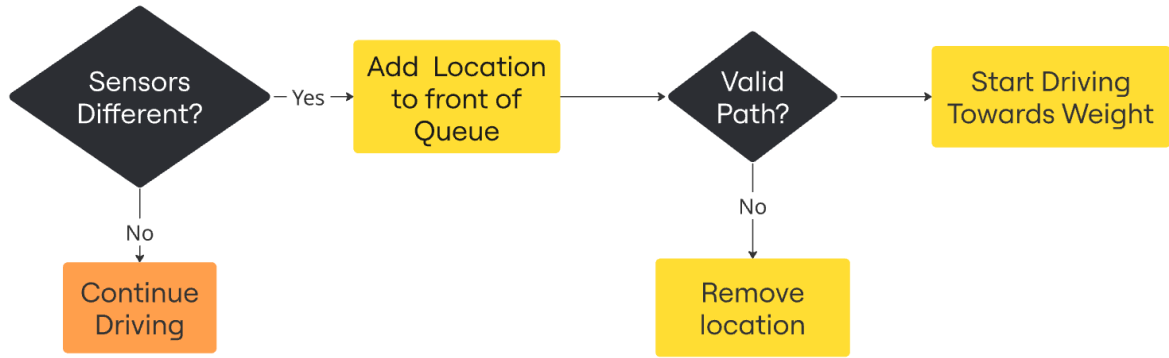


Figure 12: State diagram demonstrating how weights were detected.

If the sensors detected different values the location that was seen by the bottom sensors would be added to the front of the queue (immediately travel towards it). A path was then generated, if the path was valid it would start driving towards it, if not it would remove the location and simply continue driving on the original path.

3.6 Software Design Modifications

Between the competition day and the last report, many software features were created or added. The following are the most important modifications that had the most efficacy during testing and during competition day.

3.6.1 Updated Path Finding Algorithm

The key improvement in software came from using post processing on the generated A* path, converting it to a Theta* path. This converted what was a 45/90 degree path finding algorithm to an any angle path finding algorithm. During the A* search a list of nodes were generated for which the robot should travel to. This was then fed through an algorithm which checked each node with ones in front of it and anytime there was line of sight (no wall in between) all nodes in between would be removed. The result of the post-processing is demonstrated in Figure 13, where Figure 13a is the path generated by A* and Figure 13b is the finalised path.

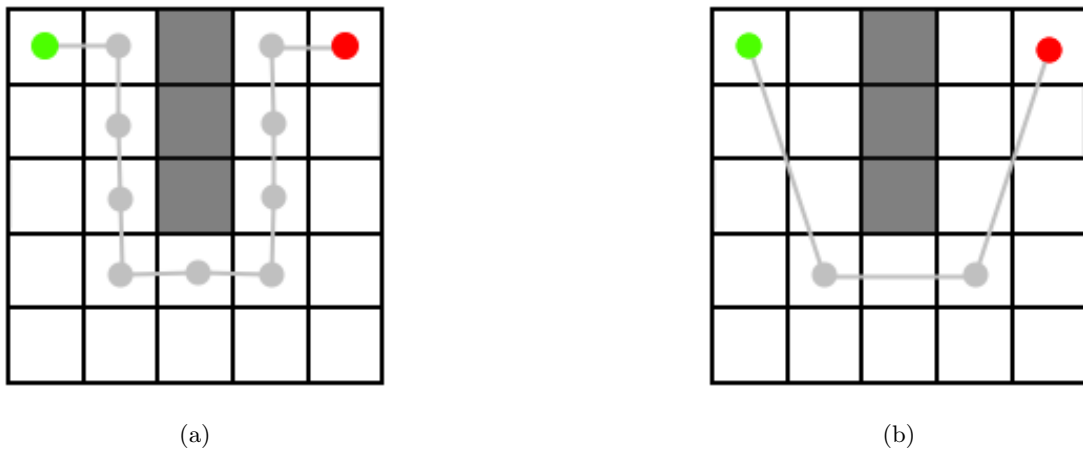


Figure 13: Original path finding algorithm A* (a) in comparison to the post processed version Theta* (b)

3.6.2 Increasing incoming map data

The secondary improvement was increasing the total data that the robot was receiving. As mentioned in the mechanical improvements section, more TOF sensors were added at the front of the robot. The map within the code was then updated exactly the same as with the initial front TOF sensor, a variable was added to the code to update a map based on an angle offset from the center. This meant that each sensor would update information in the correct direction from where it believed it was. This addition significantly improved edge cases such as when the robot would drive forward without sensing any walls because the wall was outside the range of the front sensor. This edge case is demonstrated in Figure 14 showing how the front sensor does not see the object that would hit the left of the robot if it were to drive forward.

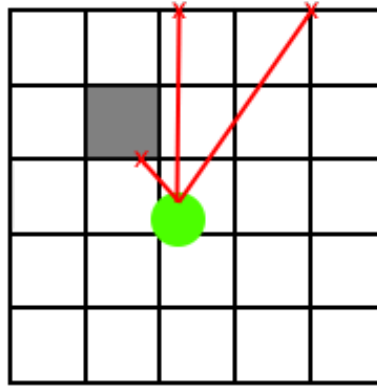


Figure 14: Additional sensors used to detect walls slightly aside from straight middle sensor.

3.6.3 Weighted Search vs Random Search

Another improvement made was changing from a random search to a weighted search algorithm. Figure 15a shows how the robot originally decided which location to travel to. This was a uniform probability between each available cell. In the example 16 cells had an even probability of 0.07 or 7% to be selected. This was improved to be weighted towards unseen cells, Figure 15a shows each cell with a reference number in the top left corner showing an example of how many times that cell had been seen by the sensor. This information was then used with a quadratic weighting function to distribute probability among the less seen cells. As shown in the Figure 15b, cells that had been seen 0 times had a 14% chance to be selected compared to cells seen 9 times having a 1% probability.

		0.07	0.07	
0.07		0.07		0.07
0.07		0.07	0.07	0.07
0.07	0.07	0.07		0.07
●	0.07	0.07		0.07

(a)

		1 0.09	0 0.14	
9 0.01		1 0.09		0 0.14
9 0.01		4 0.04	3 0.05	1 0.09
9 0.01	9 0.01	8 0.02		0 0.14
●	9 0.01	9 0.01		0 0.14

(b)

Figure 15: Original search weights (a) vs weighted search weights (b) with observation counts top left of cell

4 Results and Evaluation

4.1 Overview

Out of six rounds the Sucky Sucky 9000 won four rounds and lost two, providing a 67% win rating. The win rating was heavily dependent on the reliability of the weights entering the sorter through the ramp. During rounds, the ramp and the sorter were prone to misalignment issues, causing the weights to get jammed. This led to coin-flip situations during a round on whether the robot would operate as intended, or continue to search for weights indefinitely, filling the inside of the robot. This issue was compounded by the bad strategy call for the fastest path to weights navigation, as the weights had no chance to bounce into place from vibrations before the next weight came in, which blocked the sorter. An upgraded ramp was introduced between day one and day two of the competition; however, this delivered minimal improvements. This proved evident during round 70, where the robot attempted to pick up eight separate weights; only failing to collect the last three as the area between the harvester and the sorter was already filled in its entirety. Referencing the fault tree analysis in Appendix B, more care was needed to ensure that the weights entered using the correct orientation.

The aspects that excelled in the design was the effectiveness of the navigation, weight sensing, traversal and harvester operation which was essential to meet the requirements. Under the stress of the competition, the navigation algorithm followed the direct inputs given, and was able to adapt to external factors such as other robots within the arena and varying walls and weight placement. The navigation system updated quickly and effectively when paths were blocked or opened up as new locations were opened.

The mechanical aspects of the movement system and structure were reliable across all terrains, including the ramp and other obstructions. It was unable to get stuck in corners that many other were obstructed by, and its circular design allowed it to bounce off and move away from competitor robots during the rounds. The harvester was also extremely reliable, with the structure being able to handle abuse from competitor robots and terrain, as well as the shielding due to it being inside the bounds of the circle, ensuring it would last throughout the round. The 50 RPM motor allowed for all weights to be grabbed and sent up the ramp, and also did not suffer any electrical faults.

The table of results from all matches are shown in Table 1.

Table 1: Table of winners and points from 6 rounds participated in the Robocup competition

		Onboard				Base			
Round	Group	Onboard Score	Metal	Plastic	Snitch	Base Score	Metal	Plastic	Winner
7	28	0.75	1	0	N	0	0	0	28
	35	0	0	0	N	0	0	0	
21	28	1.25	2	0	N	0	0	0	28
	35	0.5	1	0	N	0	0	0	
40	12	0.75	1	0	N	3.5	2	1	12
	28	1.75	2	0	N	0	0	0	
57	28	2.5	0	2	Y	1	1	0	28
	14	1.25	2	0	N	0	0	0	
66	28	1.5	2	0	N	1	1	0	28
	26	0	0	0	N	0	0	0	
70	39	1.25	2	1	N	0	0	0	39
	28	0.75	1	1	N	0	0	0	

As seen in Table 1, our robot managed to function consistently overall, with a minimum score of 0.75, and a maximum score of 3.5. Our robot was able to collect an average of 2.33 weights per round, the fifth highest in the competition. These results are reflective of meeting many of the specifications met. As seen in Appendix A, the robot was able to complete almost all of the performance requirements laid out (1.1, 1.3, 1.4, 1.5), except for requirement 1.2, as even though weights could be sorted within five seconds, the jamming of the sorter meant that some weights could not be sorted consistently. In reference to the functional requirements, all motion related requirements (2.1, 2.2, 2.3) were met. The functional requirements 2.4 and 2.5 were not passed, as this could not be achieved reliably. In reference to picking up weights regardless of orientation, the harvester was capable of this, however, due to jamming, not all weights would make it to the sorter. This also effects the ability for the robot to track weights, which could only work when weights fully entered the sorter. The operational requirements (3.1) for the robot to operate for 10 minutes was successfully met. The non-functional requirements were also met (4.1), as many major parts could be and were changed leading up to the competition. Both constraints (5.1, 5.2) were met allowing the robot to meet the competition specifications.

4.2 Round 40 - Group 12

4.2.1 Performance table

Table 2: Group 12 performance characteristic table

Group 12 Performance characteristic table			
Design characteristics	Quantitative/qualitative Figures	Comparison to current design	verdict on characteristic
Weight collection	Collected 100% of all weights that entered through the harvester.	Group 12's harvester system is similar to current collection method, however it was highly refined for pickup efficiency, being effective in all matches.	EFFECTIVE
Weight detection	Approximately 70% of weights surrounding the robot were detected and approached by the design.	Group 12 utilised wider view array for weight detection that allowed for the robot to respond much faster compared to the front-focused TOF sensor design of our robot.	EFFECTIVE
Weight storage	Correctly sorted and rejected weights to an effectiveness of 80 %.	The sorting design was far more effective than the current design, with the system refined and tuned well.	EFFECTIVE
Traction system	Tracks were effective through all matches	The tracks allowed for continual locomotion around the track.	EFFECTIVE
Navigation	robot's navigation system enabled the robot to never get stuck.	Navigation systems from both robots were similar in quality, with group 12's robot being able to navigate the arena without impedance	EFFECTIVE
Wire organisation	wiring was organised and labelled	Similar quality level of wire organisation.	EFFECTIVE

4.2.2 Performance During Round

Group 12's robot performed exceptionally, placing first overall in the competition. Our robot competed against them in round 40, serving us our first loss; with Group 12 achieving a score of 4.25, the highest score obtained in a round without catching the snitch. This group's robot made use of a similar harvester collection system to our robot. The overarching design for the sorting system and navigation system followed a more simplistic nature. The sorting system relied on the use of an inductive sensor to sort between metallic and plastic, then sweeping the weight into one of two trays in the back for storage. The navigation used a comparatively less complex wall following algorithm to search for weights. Group 12's design is shown in Figure 16.

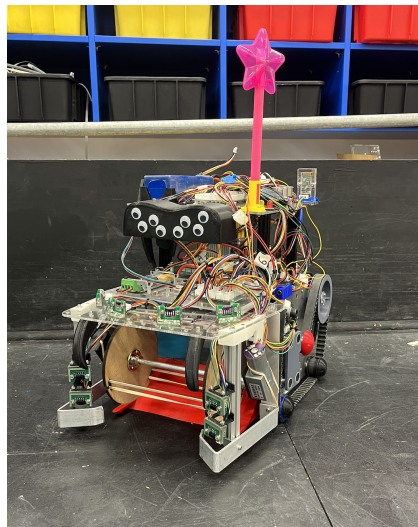


Figure 16: Group 12 robot design

4.2.3 Weight Collection

Group 12's robot used a very similar harvester system for weight collection. Their ramp into the sorting system shared a range of similarities, including side walls to guide weights into the entrance of the sorting system. Its ramp design

differed slightly, with a shorter ramp, which was much lower to the grounds, which benefited the collection system to sorting system flow. The sorting system design resembled the initial design of our robots sorting system laid out in the conceptual design report. It consisted of a stopper in the centre of the ramp to hold the unsorted weight in place, followed by a pinball type arm movement to knock the weight in the storage or out the back. A single proximity sensor was used to check whether the weight was magnetic or plastic. As seen in Table 2, Group 12 proved to be more effective in its weight collection, detection, and storage.

Group 12's robot would orient the weights as they were pushed up the ramp, allowing weights to be picked up in any orientation, increasing the overall number of weights they could collect during a round. The open mouth harvester design allowed for quick weight collection, allowing the robot to be effective both at the beginning of the round with weights in their set positions, and later in the round as weights were knocked over and moved about.

Overall, Group 12's collection and sorting system functioned in a similar way to our own design, but improved on inefficiencies present in our design, allowing for more consistent operation throughout each round. Due to our sorter's reliability issues with getting jammed on the ramp, Group 12's design in contrast showed the benefits of a design which allowed for larger tolerances, which promoted reliability in an uncontrolled environment. This left their ability to collect weights to remain effective during each round, independent of environmental changes.

4.2.4 Area Navigation & Sensor Design

Group 12 utilised a similar sensor layout for navigation. This consisted of an array of four TOF sensors on the front of the robot for navigation, as well as two sets of TOF sensors set to low and high heights to detect weights through a difference of distances measured. In addition, the robot also used two ultrasonic sensors on either side. Similar to our robot, the weight detection proved consistent at differentiating obstacles and objects. Their detection differed slightly from ours in layout, by having the sensors angled inwards to the front of the robot rather than outwards; narrowing its range of sight to detect weights. This difference appeared to provide similar results to our robot's detection.

Group 12 designed an effective goal focused algorithm which worked comparatively simply compared to our robots. Detection data was used to both identify walls with the top TOF sensor array, and identify both obstacles and weights using the two sets of TOF at the front as well as the ultrasonic sensors. The robot would move forward with no specific location, and avoid obstacles if detected. If weights were detected, moving towards these would be prioritised over avoiding obstacles. As seen in Table 2, the robots movement and navigation was as effective as our robots; allowing it to move about without impedance.

4.3 Round 66 - Group 26

4.3.1 Performance table

Table 3: Group 26 performance characteristic table

Group 26 Performance characteristic table			
Design characteristics	Quantitative Figures	ISO Comparison to current design	verdict on characteristic
Weight collection	Collected weight in 5 seconds on average. Jammed and broke after second weight.	Group 26's harvester system is similar to current collection method. Reliability issues downgraded the system	INEFFECTIVE
Weight detection	67% effective in detecting weights	ALA The weight detection system was similar to the current design, however was ineffective due to programming and sensor reliability issues	PARTIALLY EFFECTIVE
Weight storage	Weight storage did jam during the match	The weight sorting system utilising a inductive and TOF sensor to detect a weight was novel, but the system jammed and was not particularly effective	PARTIALLY EFFECTIVE
Traction system	Traction system was 100% effective, constantly moving the robot	The track system was similar to the current design and lasted considerable wear and tear	EFFECTIVE
Navigation	robot consistently got stuck in matches	The navigation system was incredible basic, utilising a wall following algorithm that did consistently break during matches and testing	INEFFECTIVE
Wire organisation	Wires were tapped down, organised and label	The wire organisation was far more effective than the current design, making sure the robot could easily be diagnosed and inspected.	EFFECTIVE

4.3.2 Performance During Round

Group 26's robot performed with mixed results, winning two of four rounds. Through this, only three weights were collected, with none of them being returned to home base. Due to a snitch catch, the robot managed to collect one weight on average, and placed eighth overall. Our robot performed well against this robot, as it proceeded to showcase everything that could go wrong. At the start of the round, group 26's robot showed promise, managing to detect and try to collect a weight early. Unfortunately, due to general wear, the grub screws holding the harvester drum to its driveshaft came loose, causing the barrel to stop spinning while the motor continued to spin continuously. This metal weight did continue to stay against the barrel for a period of time. Things got worse as the TOF sensors timed out, causing the robot to stop properly detecting walls. This resulted in the worst possible outcome, with their robot driving into our base, depositing the weight there, and then getting stuck driving forward, with the watchdog timer failing to go off. Group 26's robot design is shown in Figure 17.

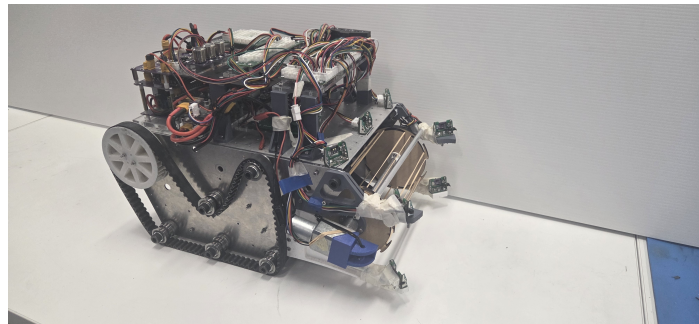


Figure 17: Group 26 robot design

4.3.3 Weight Collection

Once again, this group's robot made use of a harvester design for weight collection. This proved capable when working correctly, but suffered from reliability issues. One issue appeared while facing our robot, when the grub screws holding the driveshaft to the barrel came loose, causing the harvester to stop spinning while the driveshaft still rotated inside. This same issue was identified in our robot's testing, and was fixed by extending a grub screw into a hole drilled in the actual rod.

The sorting system proved effective enough, as seen in Table 3. This design again used an inductive proximity sensor within the robot to sort weights once they were collected. This stored weights in the same orientation as our robot's sorting, but would then either tip plastic weights sideways, or scoop metal weights into the rear of the robot.

4.3.4 Area Navigation & Sensor Design

This robot again used almost the exact same sensor layout, with the only difference being that its weight detecting TOFs were turned inwards rather than outwards. This navigation provided a good range of sight when it worked, but overall tended to struggle to differentiate between objects and open space. An added colour sensor was placed at the bottom to identify if it was moving over its base, allowing it to deposit weights. This meant that depositing weights at base was entirely due to luck; this opportunity never presented itself during the competition.

Group 26 utilised a very simple navigation algorithm. This relied on a finite state machine which would drive forward until it identified an object. If this was detected as an obstacle, it would turn away, and if it detected a weight, it would turn towards it. This performed with mixed results, sometimes failing to detect a wall causing it to drive towards it. The watchdog timer appeared to have a very large delay, likely failing to detect when it was stuck in many scenarios. It should also be noted that this navigation failed to fix the issue of the TOF sensors timing out during operation, causing the robot to effectively go partially blind at random moments during each round. Overall, the navigation was the largest issue holding this robot back.

4.4 Round 70 - Group 39

4.4.1 Performance table

Table 4: Group 39 performance characteristic table

Group 39 Performance characteristic table			
Design characteristics	Quantitative/qualitative Figures	ISO Comparison to current design	verdict on characteristic
Weight collection	Weight collection was 90% effective	Group 39's system utilised rubber wheels and a conveyor belt to guide the weights up into the sorter	EFFECTIVE
Weight detection	The robot detected and approached 80% of weights in vicinity	The weight detection was quite effective however was let down by the tractional system	PARTIALLY EFFECTIVE
Weight storage	Weight storage was 100% effective	Weights were sorted beforehand and was a simple container compared to the current design, but was quite effective	EFFECTIVE
Traction system	Traction system movement accuracy was around 65%	The tractional system was the typical variety for Robocup robots, however it suffered from slipping and coordination issues	EFFECTIVE
Navigation	The had 3 times were it got stuck in the competition	The navigation system was far more basic than the current design, and had no algorithm or backup in case it rammed the wall	PARTIALLY EFFECTIVE
Wire organisation	Wires are partially organised	AND compared to the current design there are some minor rats nests of wiring for the robot, however it is not significant	PARTIALLY EFFECTIVE

4.4.2 Performance During Round

Group 39 utilised a conveyor belt system to pull weights into it. This collection proved consistent, collecting an average of 1.60 target weights per round. Their robot was effective at holding weights without them falling out. Group 39's robot's largest strength was its ability to return to home base and drop off weights, with a total of four weights being

held onboard at the end of the competition, and three at base; giving it a total of 47% of collected weights being dropped off at home base. Group 39's robot is shown in Figure 18 below.

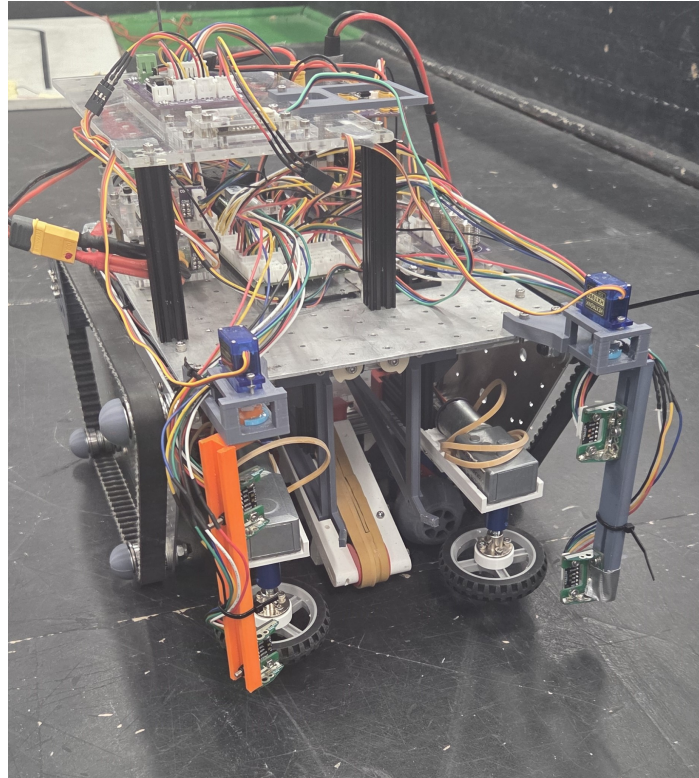


Figure 18: Group 39 robot design

4.4.3 Weight Collection

Group 39 built a highly unique sorting system, consisting of a rotating rubber belt to act as a conveyor belt system. Two horizontal free spinning wheels were used at the front to help orient the weights. Against our robot this proved to be as effective, as it had consistent reliability, could pick up weights in different orientations, and was unable to differentiate between metallic and plastic. The sorting system was much more simplistic, and would just hold all weights inside without sorting. A door was used in conjugation with a colour sensor and navigation so that it could deposit all weights at base. As identified in Table 4, this appeared to have mixed results, largely due to navigation issues. It could be said that our robots sorting was much more effective due to its ability to sort weights effectively once inside, however, since this team experienced no issues with weights entering its storage, it ended up being equally reliable.

4.4.4 Area Navigation & Sensor Design

Like the other robots, group 39 used two sets of TOF sensors at different heights to detect objects. This was adapted slightly by having the pole the sensors were attached to rotate back and forth to detect a wider area. Overall, this appeared to make a minimal difference on its weight detection. Similar to ours, this robot used a combination of encoder and imu data to navigate back to base. This design differed slightly, using two dedicated separate encoders attached to wheels in the centre of the robot. These wheels were orientated horizontally and vertically to track the robots movement in Cartesian coordinates without having to rely on the motors built in encoders. Referring to Table 4, this proved to have mixed results, as during our round it completely failed to return to base, finishing in the middle of the arena.

4.5 Post Mortem Summary

4.5.1 Geneva Mechanism

From the competition analysis, serious improvement is necessary for the weight collection system. Unfortunately, in hindsight the testing undertaken for that component was not sufficient, as it did not test for the unique edge case of the weight jamming mid-ramp. This left the robot with an under-tested and unrefined aspect when compared to the other features of the robot. The main issue that was encountered was motor slippage for the drum causing improper relocation for the weight collection. The solution would be the utilisation of a Geneva mechanism, shown below in Figure 19.

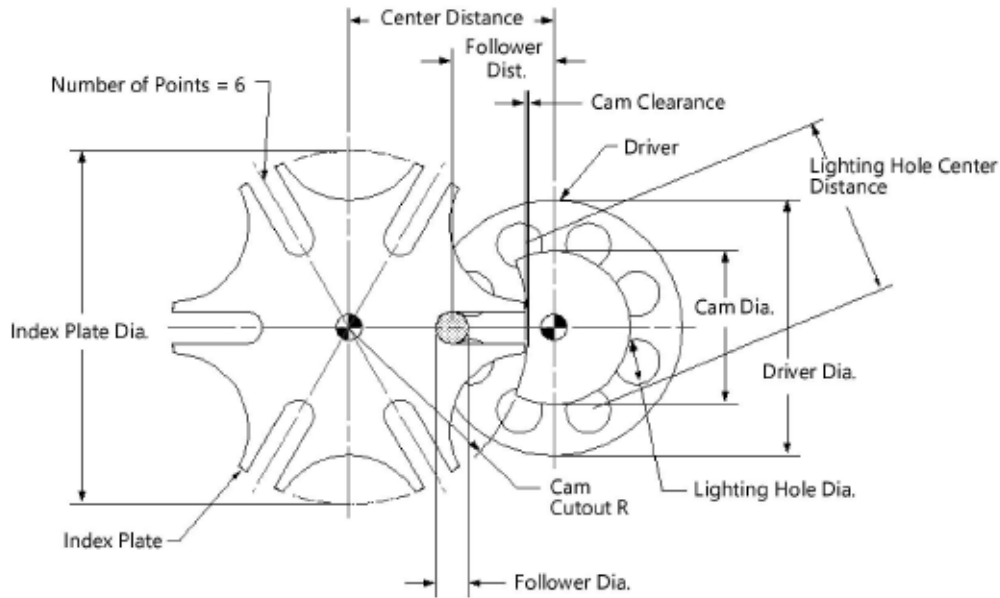


Figure 19: Geneva mechanism used for gear-like rotation

The Geneva mechanism would allow for turning the drum in specific increments and would solve the motor alignment issue for the drum. There would need to be additional space made for the system, but it would improve effectiveness of the drum.

4.5.2 Watchdog Timer

During round 66, the robot got stuck for a total of 48 seconds against the wall. This was due to the lack of a watchdog timer checking whether we were stuck or not. To significantly improve this stuck time, the IMU would be checked to determine whether or not the robot was actually moving, if not it would then try reversing from its current position. This would hopefully fix the problem of getting stuck and thinking that it was actually travelling to its target location.

5 Conclusion

The final robot design was largely fit for purpose, meeting the original performance requirements and specifications defined in Report 1. The robot adapted effectively with the competitive environment. The movement and navigation system operated reliably, with the Theta* algorithm allowing for efficient movement and object avoidance. The weight detection system successfully detected and travelled to weights within the arena, though mechanical and sensing limitations reduced consistency in collection.

The robot demonstrated strong overall system integration, balancing mechanical, electrical and software design. The main limiting factor was the mechanical jamming between the ramp and the sorting chamber. This limited the robots overall output within rounds. Additional development would be used to further improve the robot by trying to utilise Design for Manufacturing and Assembly (DfMA) and increasing testing in full arena conditions, which would of pointed out the main flaws before competition day. Another development would be refactoring the code to C which would of increased performance allowing for faster execution speed. These refinement would have enhanced the overall reliability of the system and reduced maintenance time allowing for more time spent discovering and fixing other issues.

Overall, the robot achieved its intended purpose as a fully autonomous system which both navigated, collected and sorted weights within an arena.

A Robot Updated Specifications

1. Performance Requirements

- 1.1 The robot should be able to pick up weights up to 1 kg (without need of initial discrimination)
Rationale: There is not enough available time in the match nor reliable enough design to try to discriminate the weight's mass and before pick up.
- 1.2 The robot should be able to detect a weight within 250mm of itself (without need of initial discrimination)
Rationale: The robot needs to have some guidance to weights to make it more effective at weight pickup.
- 1.3 The design must be able to determine what type of weight it has picked up in a maximum 5 second time frame
Rationale: If there are multiple weights that the robot encounters, the weight processing must be as efficient as possible.
- 1.4 The robot must be able to capture the snitch, even if that does sacrifice performance.
Rationale: The snitch is a valuable asset to capture during the competition, and its size makes it quite difficult to properly process without increased dimensional constraints.
- 1.5 The robot must be able to climb a 30 °incline and a maximum height of 100mm
Rationale: Previous competitors had large issues with ramps leading to large failures

2. Functional Requirements

- 2.1 The robot should be able to turn around in any corners without getting stuck
- 2.2 The robot shall be capable of turning 360 degrees while remaining in place
Rationale: These features allow the robot to be able to reorientate itself at any location in the arena
- 2.3 The robots' track system must be able to move the robot at walking pace (2.9km/h) when fully loaded.
Rationale: The time pressure of the match is a major factor, the faster the robot can traverse the course, the faster it can collect weights when considering competition.
- 2.4 Weights must be able to be picked up no matter orientation into the system
Rationale: During a round weights are knocked over and moved, this leads to a variety of orientations that would need to be picked up
- 2.5 The robot must keep track of the amount of weights on board the robot
Rationale: Holding more than 2 weights at the end of the round will lead to point deductions.

3. Operational Requirements

- 3.1 All components of the robot, mechanical or electrical, must be able to function for 10 minutes or more.
Rationale: The match only lasts a fraction of the time, however stress testing the robot with longer times and introducing stressful situations on the system may present deficiencies that can be fixed before the proper competition.

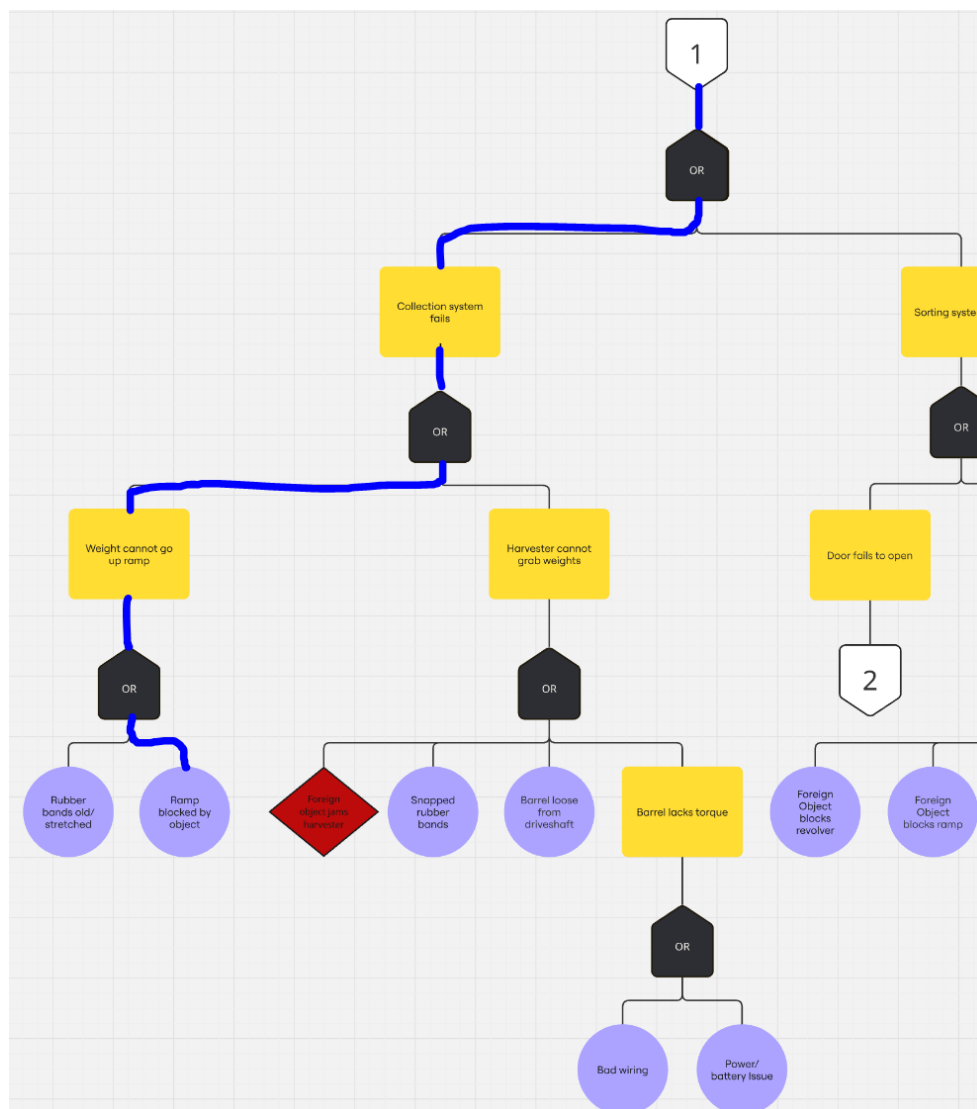
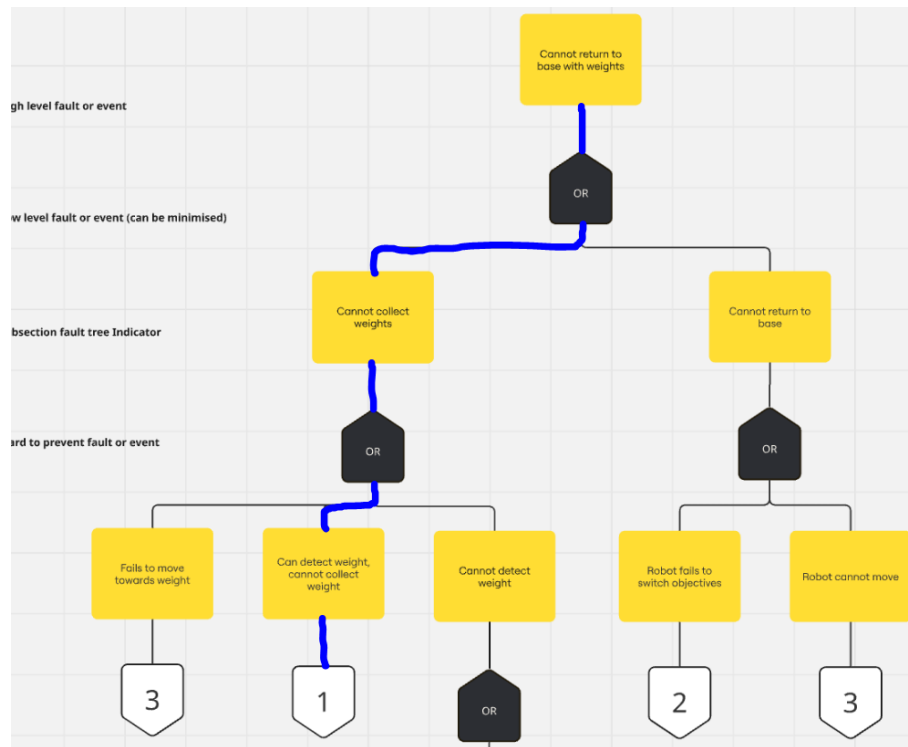
4. Non-functional Requirements

- 4.1 The robot's components must be easily removed and replaced to allow for modularity
Rationale: The time between rounds is short, if modification needs to be made it needs to be efficient and quick

5. Constraints

- 5.1 The robot must fit in a 400x400mm area.
Rationale: The tightest corners of the design are within that 400mm size, so a robot that can traverse in tight areas like that will be able to go across majority of the course fine.
- 5.2 All parts fabricated should be able to be fabricated at university with no external help or specific constraints.
Rationale: If failures occur during the competition, parts would need to be fabricated within the university.

B Fault Tree Pathway For Mechanical Failure



C Bill Of Materials

C.1 Bill Of Materials: 3D Printed Components

3D Printed Bill Of Materials			
Component Name	Number of Parts	Part Weight (g)	Total Material (g)
Sorter Drum	1	127.11	127.11
Sorter Main Body Connector	1	14.86	14.86
Sorter Motor Holder	1	17.81	17.81
Sorter Weight guide	1	11.16	11.16
Sorter Drum Cas- ing	1	137.17	137.17
Sorter Door	1	15.15	15.15
Sorter Door Motor holder	1	8.43	8.43
Structural Medium-Sized Tensioner Profile	10	11.16	111.6
Structural Small- Sized Tensioner Profile	12	4.19	50.28
Structural Large- Sized Half Ten- sioner Profile	2	11,81	23.62
Structural small- Sized Half Ten- sioner Profile	2	4.61	9.22
Structural 1-Sensor holder	4	4.68	18.72
Structural 3-Sensor holder	1	30.46	30.46
Harvester Motor Holder Left side	1	26.8	26.8
Harvester Motor Holder Right side	1	23.75	23.75
Harvester Motor Tensioner Profile Right side	1	10.88	10.88
Harvester Motor Tensioner Profile 1 Left side	1	6.65	6.65
Harvester Motor Tensioner Profile 2 Left side	1	4.38	4.38
Battery Holder	2	5.69	11.38
Harvester Ramp	1	78.22	78.22
Harvester Wall	2	51.1	102.2

3D Printed Bill Of Materials			
Component Name	Number of Parts	Part Weight	Material Cost
Drive Motor Holder	2	31.00	62.00
Motor Meshing Gear	2	18.75	37.5
Motor Meshing Gear Rod	2	3.61	7.22
Motor Meshing Gear Support	2	5.68	11.36
Tensioner Profile	4	5.23	20.92
Wire sorter	2	2.86	5.72
TOTAL GRAM-MAGE	——	——	1017.38

C.2 Bill Of Materials: Pre-Fabricated Components

Pre-Fabricated Bill of Materials			
Component Name	Number of Parts	part Cost	Total Cost
DC Motor — 28PA51G	2	70.0	140
DC Motor — JGY-370	1	22.0	22.0
DC Motor — SKU505979	1	15.0	15.0
Standard servo — HX12K	1	14.0	14.0
Open beam Aluminum profile	4	-	-
robot track	2	-	-
Round Bar Aluminium 8mm	2	-	-
Tennsy 4.0	1	-	-
Stop Go button	1	-	-
Power Supply Board	1	-	-
PPM Motor Driver Board	2	20.0	40.0
Digital Level Shift IO Board	1	-	-
Encoder IO Board	1	-	-
Inductor Level Shift IO Board	1	-	-
TOF I2C — VL53L1XV2	5	10.0	50.0
TOF I2C — VL53L0XV2	2	5.0	10.0
IMU — SEN0253	1	36.0	36.0
Inductive Proximity	1	25.0	25.0
Limit Switch	2	-	-
Drive track support hardware	6	03.0	18.0
Rigid Flange Coupling	3	01.8	5.4
Flanged bearings	3	01.0	03.0
Battery	1	-	-